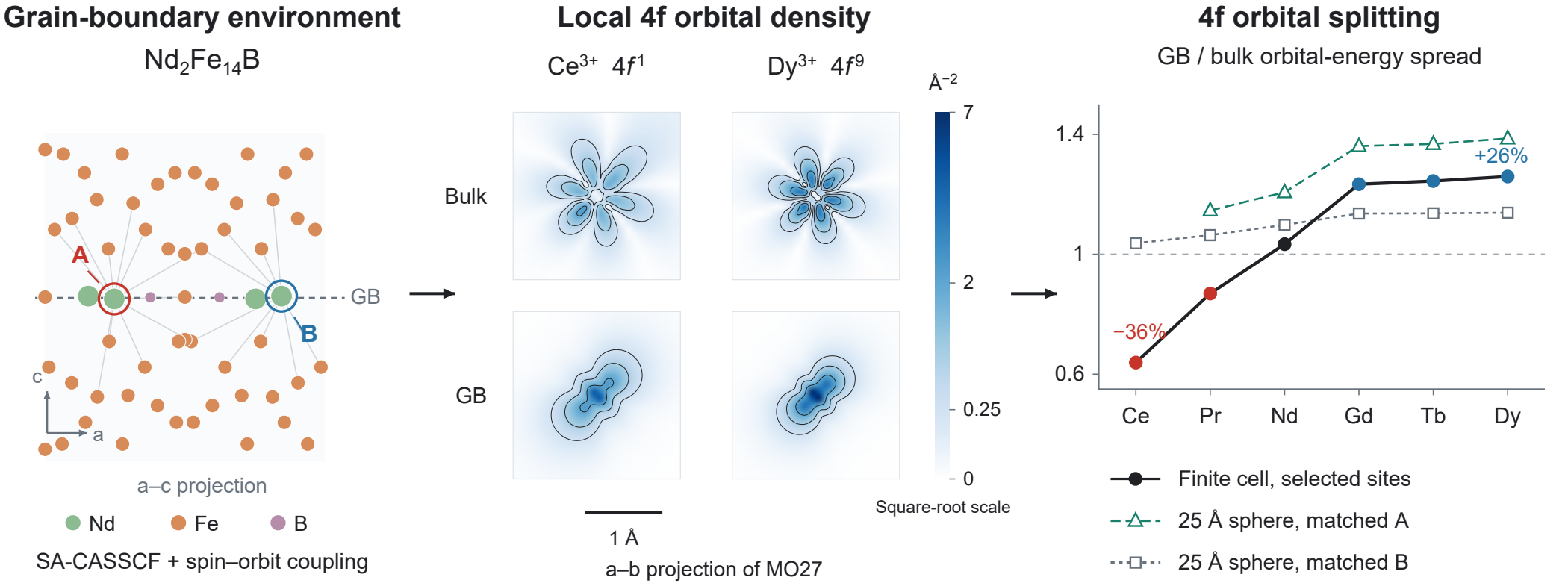

Grain-boundary environment
$Nd_2Fe_{14}B$
A
B
GB
c
a
a–c projection
Nd
Fe
B
SA-CASSCF + spin–orbit coupling
Local 4f orbital density
Ce^{3+} $4f^1$
Dy^{3+} $4f^9$
Bulk
GB
Å^{-2}
7
2
0.25
0
Square-root scale
1 Å
a–b projection of MO27
4f orbital splitting
GB / bulk orbital-energy spread
1.4
1
0.6
−36%
+26%
Ce
Pr
Nd
Gd
Tb
Dy
Finite cell, selected sites
25 Å sphere, matched A
25 Å sphere, matched B